

BUSINESS DECISION

Can a rate increase pay for the renewals it loses?

RECOMMENDATION

Investigate a bounded +5% option; obtain actual premium and renewal evidence before a rollout.

At assumed elasticity 1.2, annual contribution increases by EUR 1.02m, with 6,537 fewer retained policies. At elasticity 6, contribution falls by EUR 0.30m. These are conditional simulations, not achieved savings.

What the rate experiment shows

Rate / response	Retained	Contribution	Change
No change	114,959	5.27	+0.00
+5%; elasticity 1.2	108,422	6.28	+1.02
+5%; elasticity 3	99,306	5.76	+0.49
+5%; elasticity 6	85,784	4.97	-0.30

Contribution and change: EUR millions. Retained volume: expected policies, rounded.

The profit benefit depends on price response



Assumed annual renewal economics

135,246 policies; one renewal opportunity each; 85% baseline retention; boosting claims model; fixed GLM synthetic price anchor. Expenses: 25% of premium plus EUR 30 per retained policy. Price margin: 10%; no inflation in the table. Price = (GLM annual model loss + EUR 30) / 0.65.

With 10% claims inflation, the +5% case contributes EUR 4.48m. Its +EUR 1.12m rate benefit compares with a baseline under the same inflation. Current premiums and observed renewals are unavailable; response and prices are assumed.

EVIDENCE, JUDGEMENT AND NEXT ACTION

Better risk differentiation. A measured path to action.

Evidence from a frozen holdout

Historical test measure	GLM	Boosting
Prediction error: lower is better	76.35	73.88
Risk ranking (Gini): higher is better	0.359	0.423
Actual / expected cost: target 1	0.886	0.961

Boosting reduces recorded-cost prediction error by 3.24%. The paired 95% sampling range for its error difference is -5.74 to -0.39, favouring boosting. Gini improvement is supported; top-decile lift improvement remains uncertain. Error uses exposure-weighted Tweedie deviance (power 1.5).

How the model was built

Public freMTPL2 data: 678,013 policies and 26,444 linked claims. Poisson GLM predicts annual claim frequency using exposure; Gamma GLM predicts individual claim size. Their product gives annual recorded loss cost. Boosting models the same components with nonlinear effects. Training, selection and final reporting use separate policy splits (60% / 20% / 20%); claims stay with their policy.

Model choice and review effort

Use boosting for illustrative scenario exploration and retain GLM as the transparent benchmark. GLM factors are easier to explain; boosting improves measured fit but needs more stability and explanation review. Familiarity does not establish regulatory acceptance for either model. This project has no production or regulatory approval.

What limits the decision

9,116 policies report claims with no cost record; those are not established zero losses. The largest 1% of linked claims account for about 38% of cost. Incomplete development, future drift and fitting uncertainty are outside the reported sampling intervals. Age 75+ GLM actual/expected reverses from about 1.67 on validation to 0.70 on test: do not turn an isolated segment ratio into a rate rule.

Proposed next decision

Obtain developed claims, current premiums, renewal outcomes and expense economics. Validate over time and customers; review permitted variables, fairness and stability. Only then assess a bounded pilot with agreed contribution and retention limits. Monitor actual vs expected claims and renewal response, with a documented rollback decision.

Dashboard: change a rate, stress elasticity, compare models at one price anchor, and save the scenario. Contribution excludes capital, reinsurance, tax and investment income. Historical French EUR results do not establish current Australian prices.

Source, dashboard and full evidence: github.com/TanmaySomani/Insurance-pricing-calculator

Sources: OpenML 41214 / 41215 v1; committed metrics, paired intervals and example scenarios. Test: 135,246 policies / 5,270 linked claims; 250 paired policy-bootstrap draws, fixed predictions.